

VerifyShelf

Real-Time Brand Protection & MAP Violation Detection

Powered by Torch Proxies Infrastructure

Key Metric	Target Value
Total Addressable Market (TAM)	\$4.2 Billion
Marketplaces Monitored	6 (Amazon, Flipkart, Daraz, Lazada, Tokopedia, Shopee)
ML Models Deployed	3 (XGBoost, Sentence-Transformers, GPT-4o)
Pricing Architecture	\$150 – \$600 per brand / month
Regional Growth Rate (CAGR)	22% — South & Southeast Asia
Active Consumer Base Served	700M+ across target platforms

Introduction

The global e-commerce landscape is undergoing rapid structural transformation, particularly across South and Southeast Asia, where marketplace platforms such as Daraz, Lazada, Tokopedia, Shopee, and Flipkart have collectively created one of the world's most dynamic and complex retail ecosystems. For brand manufacturers operating across these fragmented channels, this expansion represents an enormous commercial opportunity — but it comes with a deeply consequential operational blind spot.

When a brand's products reach third-party resellers on these platforms, the brand loses visibility. Resellers operating outside the boundaries of contractual Minimum Advertised Price (MAP) agreements undercut pricing floors, erode brand equity, and trigger cascading price wars that damage the entire distribution channel. Grey-market imports, unauthorized seller accounts, and high-fidelity counterfeit listings compound this problem — all operating invisibly, twenty-four hours a day, seven days a week.

The existing class of brand-protection tools has consistently failed to serve this market. Enterprise suites designed for Western platforms such as Amazon, eBay, and Walmart charge between \$10,000 and \$50,000 annually, pricing out the vast Small and Medium Business (SMB) segment. Critically, none of these platforms cover the fast-growing Asian marketplaces where the actual growth is happening. And without access to geo-targeted, residential-IP-based data collection, the information they provide is technically invalid — platforms like Daraz Pakistan serve different pricing to different geographic audiences, making data-center-based scraping fundamentally inaccurate.

VerifyShelf is built to close this gap. Designed on a five-layer automated intelligence pipeline and powered at its foundation by Torch Proxies' residential and ISP proxy infrastructure, VerifyShelf delivers real-time MAP violation detection, cross-marketplace seller fingerprinting, and AI-generated enforcement documentation — at an accessible \$150 to \$600 per month price point that makes professional brand protection viable for SMB brands scaling across Asia for the first time.

This document presents the complete technical architecture, market rationale, machine learning design, and go-to-market strategy for VerifyShelf as submitted to the Proxy Maze '26 Innovation Challenge. It draws directly on publicly available market research, open-source technical documentation, and validated infrastructure benchmarks to substantiate every design decision outlined herein.

1. Executive Summary

VerifyShelf is an intelligent, real-time brand protection and Minimum Advertised Price (MAP) violation detection platform engineered specifically for brand manufacturers navigating highly fragmented third-party reseller ecosystems across global e-commerce marketplaces. The platform combines geo-targeted web crawling, machine learning classification, cross-marketplace identity fingerprinting, and AI-generated legal enforcement into a single, cohesive workflow — removing the manual overhead that has historically made brand monitoring impractical for mid-market companies.

The Core Problem

Brands allocate substantial capital toward defining regional pricing strategies. However, the moment products enter decentralized marketplace channels such as Amazon, Daraz, Lazada, Flipkart, or Tokopedia, brands lose direct visibility. Unauthorized merchants undercut MAP thresholds, grey-market imports cannibalize legitimate local distributors, and counterfeit listings erode customer trust. These activities operate continuously, entirely outside the brand's direct line of sight.

The damage compounds over time. A single seller listing below MAP triggers automated pricing algorithms used by competing resellers, which respond by matching or beating the discounted price within hours. Within days, a brand's entire regional pricing architecture can collapse — not due to any deliberate campaign, but from a single unchecked violation that was never detected in time.

The Market Gap

Legacy enterprise brand-protection suites require multi-year contracts costing \$10,000 to \$50,000 annually, pricing out the vast SMB segment. Crucially, these platforms ignore emerging high-velocity marketplaces — specifically Daraz, Lazada, Tokopedia, and Shopee — which collectively serve over 700 million active consumers across South and Southeast Asia. VerifyShelf disrupts this paradigm by entering the market at an accessible \$150 to \$600 per month price point, specifically targeting the underserved SMB brand layer.

Infrastructure Dependency

VerifyShelf is structurally impossible to construct without utilizing Torch Proxies' residential and ISP proxy networks. Modern localized e-commerce platforms deploy strict anti-scraping countermeasures and serve dynamic, geo-fenced pricing matrices. Daraz Pakistan, for instance, displays localized promotional pricing exclusively to domestic residential IP addresses while flagging and blocking data center traffic in real time.

By embedding Torch Proxies' Proxy API directly into the platform architecture, VerifyShelf provisions isolated, country-specific residential routing channels for every tracking agent — transforming proxy infrastructure from a commodity utility into the platform's primary operational moat.

2. The Problem in Detail

2.1 What Is MAP Enforcement?

Minimum Advertised Price (MAP) is a contractually mandated pricing floor established by brand manufacturers to preserve product equity and protect wholesale margins across distribution channels. Resellers contractually agree never to advertise products below this floor. Despite these agreements, violations are both common and structurally difficult to detect at scale. The following mechanisms drive systematic MAP failures:

- **Grey-Market Arbitrage** — *Resellers bypass official distribution networks, source excess inventory from cheaper foreign jurisdictions, and undercut regional pricing floors.*
- **Anonymized Merchant Networks** — *Bad actors create multiple burner storefront accounts to offload unauthorized stock, evading legal penalties through corporate obfuscation.*
- **Cascading Algorithmic Price Wars** — *Automated pricing bots detect a single undercutting seller and instantly adjust competing prices downward, destroying a brand's regional pricing ecosystem within hours.*
- **Counterfeit Listings** — *High-fidelity counterfeit products are listed alongside authentic inventory, eroding consumer trust and cannibalizing legitimate sales volume.*

2.2 Marketplace Failure Matrix

The existing MAP monitoring ecosystem has five critical structural failures that VerifyShelf is specifically designed to resolve:

Failure Mechanism	Direct Brand Impact
Data Center IP Flagging	Scraping infrastructure is quickly banned, yielding stale, inaccurate, or incomplete price points that brands cannot act on.
Emerging Market Blind Spots	Zero legacy tools monitor South and Southeast Asian platforms (Daraz, Lazada, Tokopedia, Shopee), where growth is fastest.
Enterprise Cost Barriers	Mid-market and SMB brands are completely priced out of professional tools, leaving them defenseless against unauthorized sellers.
Unstructured Alert Dumps	Brand managers receive thousands of un-vetted alerts daily without severity ranking, context, or clear action items.
Siloed Identity Monitoring	Malicious sellers change storefront names to escape enforcement penalties, recurring infinitely without cross-platform identity referencing.

3. The Solution — VerifyShelf

3.1 Platform Overview

VerifyShelf executes its data ingestion, evaluation, and enforcement workflow across five automated structural layers. Each layer is designed to build on the intelligence produced by the previous stage, transforming raw marketplace data into actionable legal enforcement documentation with minimal human intervention.

Layer 1: Brand Onboarding & Provisioning | *Real-time*

Brand managers upload their product catalogs, official SKUs, MAP thresholds, and authorized distributor list via a Next.js dashboard. The backend immediately queries the Torch Proxy API to programmatically provision a dedicated sub-account for that brand workspace, isolating its bandwidth consumption, request allocations, and billing telemetry from all other clients on the platform. This ensures complete data segregation and enables per-brand proxy health monitoring.

Layer 2: Geo-Targeted Crawling Matrix | *Configurable — 1 to 6 Hour Intervals*

Celery workers trigger targeted crawling tasks orchestrated via Redis queues. Every request is routed through a localized Torch Proxies residential node mapped directly to the target marketplace's primary consumer base — Daraz Pakistan traffic routes through a Pakistani residential IP, Lazada Thailand routes through a Thai node, and so on. This architecture

captures real-time localized pricing, flash promotions, and delivery configurations that are completely invisible from outside the native region.

Layer 3: Machine Learning Violation Classification | *Under 500ms per listing*

Scraped payload data is parsed and processed through a trained XGBoost multi-class classification engine. The model assesses listing characteristics — price variance, merchant account age, listing text vectors, and image hash structure — to output an instant classification: Compliant, MAP Violation, Grey Market, or Counterfeit Risk, with an associated confidence probability score.

Layer 4: Cross-Marketplace Seller Fingerprinting | *Batch asynchronous — Daily*

The data pipeline extracts merchant behavioral metadata and transforms text and image footprints into 384-dimensional dense vectors using sentence-transformers. These profiles are mapped with an HDBSCAN clustering model to match unknown merchant identities across different storefront aliases and platforms. Sellers with a cosine similarity above 0.87 to a known offender cluster are flagged as a probable same-actor re-registration — preventing bad actors from hiding behind new account names.

Layer 5: Automated Legal Enforcement & Analytics | *Immediate on approval*

When a violation is verified, structured violation context is fed into GPT-4o. The model instantly drafts platform-specific takedown requests and cease-and-desist documentation containing the violation URL, local screenshots, and policy excerpts. Brand managers review and transmit with a single click. Every Monday, a full brand health report is auto-generated in plain English — written as a human analyst narrative, not a raw data export.

4. Technical Architecture

4.1 Comprehensive Technology Stack

Layer	Component Selection & Rationale
Frontend	Next.js 14 (App Router), Tailwind CSS, shadcn/ui, Recharts, React Query — deployed via Vercel for global CDN edge performance
Backend API	FastAPI (Python 3.12), Pydantic v2, PyJWT authentication, Stripe billing infrastructure
Task Queue	Celery 5, Redis 7 (message broker and cache), Celery Beat for scheduling, Flower dashboard for monitoring
Extraction Engine	Playwright (Headless Chromium for JS rendering), HTTPX + BeautifulSoup4 for lightweight extraction

Proxy Gateway	Torch Proxy API — Dynamic Residential Geo-Targeting + Persisted ISP Proxy Pools per brand sub-account
Data Architecture	PostgreSQL 16 with pgvector for identity clusters; TimescaleDB hypertable for price time-series optimization
Object Storage	AWS S3 — raw HTML DOM archives, PDF enforcement logs, screenshot evidence assets
AI/ML Layer	XGBoost classifier, sentence-transformers all-MiniLM-L6-v2, HDBSCAN clustering, OpenAI GPT-4o API

4.2 Torch Proxies Integration — The Core Data Layer

The following structural implementation demonstrates how VerifyShelf dynamically integrates Torch Proxies' API sub-accounts, rotates geo-targeted residential credentials, and executes anti-fingerprinting crawls using Playwright. This is representative of the proxy orchestration logic that underpins every scrape in the platform:

```
proxy_host = "gate.torchproxies.com:8000"    proxy_user =
f"user-{torch_sub_id}-country-{country_code}"    browser = await
p.chromium.launch(headless=True, proxy={"server": proxy_server})    locale =
f"{country_code.lower()}-{country_code.upper()}"
```

Each brand workspace is provisioned with an isolated Torch Proxies sub-account, enabling granular per-client billing, usage telemetry, and complete isolation of proxy credentials. This sub-account architecture ensures that a single client's scrape behavior cannot affect another brand's proxy health scores or request quotas.

5. Machine Learning Model Design

5.1 Model 1 — Multi-Class Violation Classifier

Architecture: XGBoost Multi-Class Gradient-Boosted Decision Trees

Output Classes: `compliant` | `map_violation` | `grey_market` | `counterfeit_risk`

The classifier is trained on a rich feature vector derived from three signal categories:

- Pricing Vectors — *price_delta_pct* (primary signal — percentage deviation below MAP), *price_drift* over a 90-day trend window
- Merchant Signals — *account_age_days*, *review_velocity* (first-derivative monthly accumulation rate), *historical_violation_count*
- Visual and Text Footprints — *listing_title_similarity* (cosine TF-IDF distance vs. official naming), *image_hash_distance* (perceptual Hamming distance vs. authorized catalog), *shipping_origin_country*

Training strategy: seed dataset with manually labelled examples from known MAP-violating sellers, followed by active learning for ambiguous cases. Pre-launch target: F1 score above 0.91 on the violation class.

5.2 Model 2 — Cross-Marketplace Seller Fingerprinter

Architecture: Sentence-Transformers (all-MiniLM-L6-v2) + HDBSCAN Density-Based Clustering

Merchant entities are processed through semantic text models using description layout patterns, localized formatting styles, punctuation distributions, price-point rounding tactics, and listing structure. Data footprints are mapped into dense 384-dimensional vector space using pgvector within PostgreSQL.

HDBSCAN isolates and updates merchant cluster nodes asynchronously. If a newly registered merchant shows cosine similarity above 0.87 relative to an established offender cluster, the platform flags the profile — linking its cross-marketplace identity even when it operates under an entirely different business name. This feature makes VerifyShelf structurally sticky: the longer a brand uses the platform, the more accurate fingerprinting becomes for their specific repeat offenders.

5.3 Model 3 — Context-Injected Automated Legal Drafter

Architecture: GPT-4o fine-tuned via Low-Rank Adaptation (LoRA) for statutory document processing

The model acts on verified violations by processing structured JSON metadata containing the product's official registration files, active violation metrics, and target merchant profiles. It outputs platform-specific cease-and-desist forms and takedown requests tailored to each marketplace's enforcement language requirements — avoiding generic boilerplate templates that result in slower enforcement response times.

Weekly brand health reports are generated using a structured context window per brand, covering: total violations versus the prior week, the top five offending sellers with listing URLs, marketplaces with the highest violation rates, price drift trend over 90 days, and repeat offender counts. Output is rendered to PDF via Puppeteer and delivered to the brand dashboard every Monday morning automatically.

6. Database Schema

The platform maintains six core relational tables across PostgreSQL 16 and TimescaleDB, designed for both transactional accuracy and high-speed analytical queries across price time series data.

6.1 Relational Architecture (PostgreSQL 16 + pgvector)

Table	Field	Description
brands	id (UUID, PK)	Primary brand identifier
	torch_sub_id	Unique Torch API sub-account mapping token — one per brand workspace
	plan	Subscription tier: Starter / Growth / Enterprise
	crawl_frequency_hrs	Crawl interval in hours determined by plan tier
products	id (UUID, PK)	Primary product identifier
	map_price_usd	Normalized MAP price floor in USD
	official_image_hash	Perceptual hash of the authorized product image for counterfeit detection
sellers	id (UUID, PK)	Primary seller identifier
	embedding VECTOR(384)	pgvector 384-dim embedding for cross-marketplace identity matching
	risk_score	Composite 0–1 risk rating, updated after each fingerprinting batch run
listings	listed_price_usd	Normalized listing price captured at exact scrape time
	scraped_at	Timestamp with timezone of data capture event
violations	type	map_violation / grey_market / counterfeit / unauthorized
	classifier_confidence	XGBoost output probability (0.00 – 1.00)
	status	detected / under_review / letter_sent / resolved

6.2 Time-Series Architecture (TimescaleDB Hypertable)

The price_snapshots table is explicitly structured as a TimescaleDB hypertable partitioned at 7-day intervals. This setup enables sub-millisecond query execution across 90-day pricing trends while isolating older analytical data for automatic compression. Key fields include:

- time (TIMESTAMPTZ, Partition Key) — *Primary temporal index driving partition pruning across all range queries*
- min_price_usd / avg_price_usd — *Aggregated price floor and mean across all active listings for a given product-marketplace-country combination*
- violation_count (INT) — *Running violation count per partition for fast trend aggregation in weekly reports*

7. Market Opportunity

7.1 Regional Growth Vectors

The e-commerce landscape across South and Southeast Asia is expanding at a 22% compound annual growth rate — the fastest sustained growth rate of any retail segment in the world. Regional platforms such as Daraz, Lazada, Tokopedia, Shopee, and Flipkart collectively serve an aggregate consumer base exceeding 700 million active shoppers. Despite this velocity, Western compliance platforms maintain an exclusive focus on North American and European infrastructure hubs. VerifyShelf captures an entirely uncontested market segment.

- \$4.2 billion total addressable market for brand protection software globally
- 22% CAGR in South and Southeast Asian e-commerce — fastest growing retail segment worldwide
- 700M+ active consumers on regional platforms with zero current MAP enforcement tooling
- \$10,000 to \$50,000 per year enterprise pricing locks out the SMB majority entirely
- AI-generated enforcement documentation reduces brand manager workload by an estimated 8+ hours per week

7.2 Target Customer Profile

Any brand that sells through resellers on online marketplaces is a potential customer — including cosmetics, consumer electronics, fashion, health supplements, sporting goods, baby products, and consumer tech accessories. The ideal early customer is a South Asian brand already selling on Daraz or Flipkart with between 20 and 200 active SKUs and 5 to 50 resellers. As a Sri Lanka-based team, VerifyShelf has direct access to this customer segment for early acquisition, rapid iteration, and case study development under real market conditions.

7.3 Revenue Architecture

Plan	Monthly Price	Included Features
Starter	\$150 / mo	Up to 50 SKUs, 3 marketplaces, 6-hour crawl frequency, email alerts, weekly AI brand health report
Growth	\$350 / mo	Up to 200 SKUs, 6 marketplaces, 3-hour crawl, seller fingerprinting, enforcement letter generation, Slack alerts
Enterprise	\$600 / mo	Unlimited SKUs, all marketplaces, 1-hour crawl, dedicated account manager, custom report branding, full API access

8. Development Timeline & Milestones

The platform is organized across a structured 14-week execution cycle synchronized with the Proxy Maze '26 hackathon tracking guidelines. Each phase delivers independently deployable functionality before the next begins.

Weeks	Milestone Deliverables
W1 – W3	Core Backend Infrastructure & Torch Proxy Integration — FastAPI scaffold, Celery and Redis setup, sub-account provisioning pipeline, initial marketplace connection testing
W4 – W6	Playwright Scraper Arrays & Database Schema Rollout — multi-marketplace crawlers, PostgreSQL and TimescaleDB schema deployment, pgvector configuration and indexing
W7 – W9	XGBoost Classifier Tuning & Seller Fingerprinting — model training pipeline, HDBSCAN cluster build, cosine similarity threshold calibration across test sellers
W10 – W12	Next.js Dashboard Development & GPT-4o Enforcement Pipelines — brand manager console, violation feed UI, auto-generated enforcement letters, PDF report rendering
W13 – W14	Closed Beta Staging & Proxy Maze '26 Demo Finalization — live product demonstration, beta customer onboarding, final performance benchmarking and review

9. Risk Management & Structural Mitigations

Dynamic Layout Shifts

Major marketplaces modify their frontend class signatures frequently to disrupt scraper configurations, invalidating CSS-selector-based extraction pipelines without warning.

Mitigation: VerifyShelf minimizes reliance on fragile CSS selectors by extracting inner HTML directly and feeding raw DOM structures into LLM-assisted heuristic sanitizers. This makes scraper logic resilient to arbitrary front-end redesigns.

Adaptive Proxy IP Bans

Advanced platform firewalls detect automated browsing profiles and block active proxy addresses over extended sessions, gradually degrading data coverage quality.

Mitigation: The scraping orchestrator uses Torch Proxies' residential rotating nodes alongside custom anti-fingerprinting browser footprints. By varying headers, user agents,

and viewport parameters on every sequential request, the platform mimics organic human traffic patterns closely enough to avoid automated detection.

False Positive Compliance Triggers

Authorized temporary brand promotions — official flash sales, regional festival events — can be misclassified as malicious MAP violations, creating noise and eroding brand manager trust in the platform.

Mitigation: The brand dashboard includes an authorized promotional calendar. Brand managers can schedule regional pricing rule overrides, which automatically updates the classifier's exclusion logic for valid promotional windows, preventing false positive escalation during sanctioned events.

ML Model Accuracy Degradation Over Time

Seller behavior evolves as bad actors identify detection patterns and adapt their listing strategies to evade classification.

Mitigation: The XGBoost classifier is retrained on a monthly rolling window of confirmed violation data. Active learning loops flag low-confidence classifications for human review before adding them to the training corpus, ensuring the model continuously adapts to evolving seller tactics.

10. Operational Strategy & Team Strengths

VerifyShelf is engineered by a specialized technical team with direct ties to the South Asian e-commerce ecosystem. Operating out of Sri Lanka provides a clear strategic advantage: the engineering team can test, deploy, and iterate on edge-case scenarios natively on localized platforms like Daraz and regional Amazon nodes during the development phase. This native access enables real-market iteration at a speed and authenticity that no Western-based competitor can replicate.

The go-to-market strategy focuses on mid-market consumer electronics, direct-to-consumer cosmetics, and high-volume apparel brands currently scaling across regional logistics hubs in South and Southeast Asia. By demonstrating measurable cost-recovery through automated MAP compliance enforcement within the first 30 days of deployment, VerifyShelf makes customer acquisition and payback period demonstration straightforward for sales conversations.

Post-launch retention is structurally reinforced by the seller fingerprinting model: the longer a brand uses VerifyShelf, the more accurate cross-marketplace offender detection becomes for their specific distribution channels. This creates a compounding data advantage that increases switching costs over time and deepens the platform's value proposition with each enforcement cycle completed.

11. Closing Statement

VerifyShelf addresses a critical, costly, and structurally overlooked operational problem: mid-market and SMB consumer brands scaling across South and Southeast Asia currently have no affordable, automated mechanism to defend their pricing integrity, identify unauthorized channels, and protect their brand equity in the world's fastest-growing e-commerce markets.

Left unmonitored, unauthorized pricing drops rapidly destroy distributor trust, trigger algorithm-driven price collapses, and permanently damage a brand's regional market position — often before any human reviewer is even aware a violation occurred. The cost of inaction compounds silently over time.

Torch Proxies' infrastructure is not a complementary utility within VerifyShelf's product architecture — it is the foundational layer of the entire system. Every price metric, violation classification, and automated legal action relies completely on the data accuracy that geo-fenced residential routing enables. Without it, scraped prices are inaccurate, marketplace coverage is impossible, and the product does not function as described.

By building directly on Torch Proxies' proxy fabric, VerifyShelf delivers an accessible, production-grade brand protection platform built for the world's fastest-growing e-commerce markets. This is not a temporary hackathon prototype — it is a commercial platform being launched with the Proxy Maze '26 infrastructure, mentorship, and prize pool as its launchpad, with full commercial operations continuing after the Week 14 demonstration.

References

The following sources informed the technical design, market data, infrastructure choices, and competitive landscape analysis detailed in this proposal. All links were verified as active at time of submission.

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